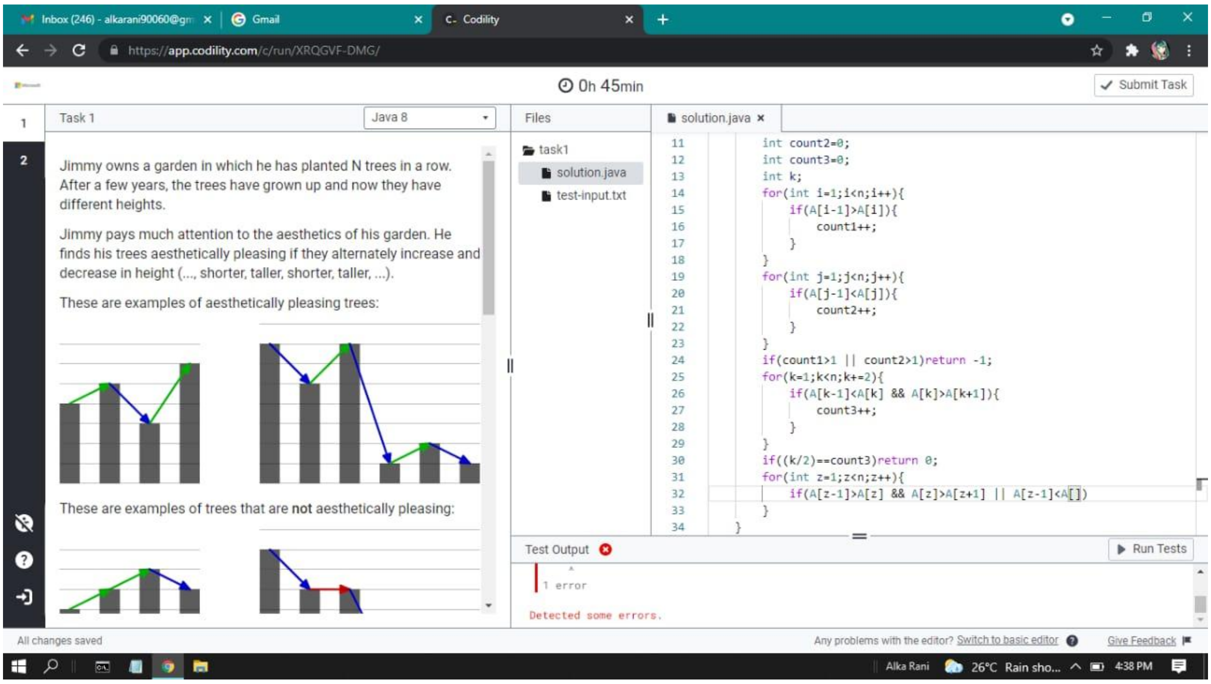
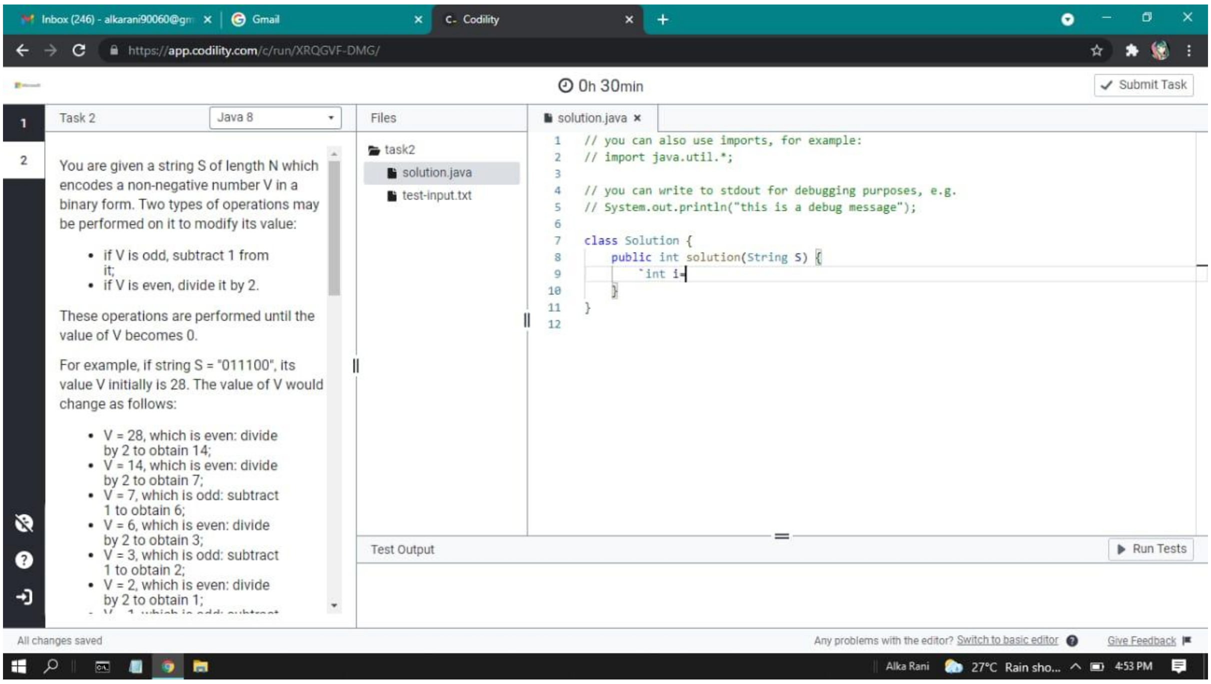
1. Microsoft I

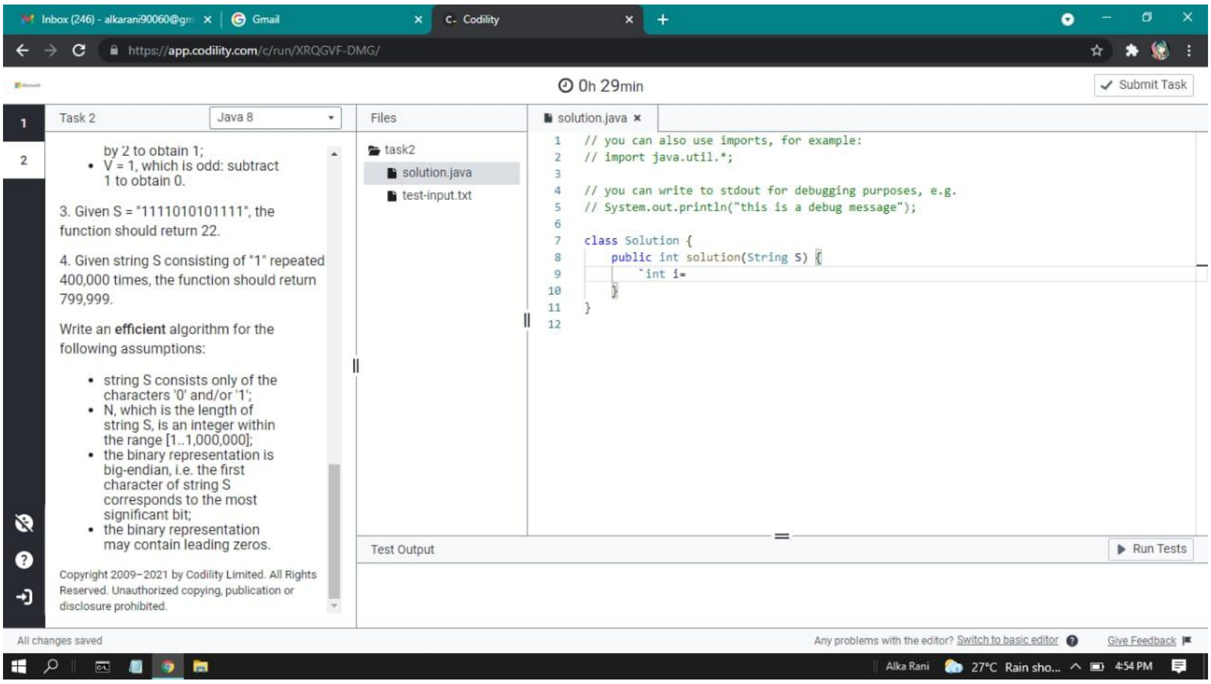
Solutions:

1. Aesthetics: https://pastebin.com/6jU2e2Zg
2. Binary Encodes: https://pastebin.com/SvKFfrhY



Solution 1:


Inbox (246) - alkarani90060@gm
Gmail
Codility
https://app.codility.com/c/run/XRQGVF-DMG/
0h 30min
Submit Task
Task 2
Java 8
You are given a string S of length N which encodes a non-negative number V in a binary form. Two types of operations may be performed on it to modify its value:
• if V is odd, subtract 1 from it;
• if V is even, divide it by 2.
These operations are performed until the value of V becomes 0.
For example, if string S = "011100", its value V initially is 28. The value of V would change as follows:
• V = 28, which is even: divide by 2 to obtain 14;
• V = 14, which is even: divide by 2 to obtain 7;
• V = 7, which is odd: subtract 1 to obtain 6;
• V = 6, which is even: divide by 2 to obtain 3;
• V = 3, which is odd: subtract 1 to obtain 2;
• V = 2, which is even: divide by 2 to obtain 1;
Files
task2
solution.java
test-input.txt
// you can also use imports, for example:
// import java.util.*;
// you can write to stdout for debugging purposes, e.g.
// System.out.println("this is a debug message");
class Solution {
public int solution(String S) {
`int i=
}
}
Test Output
Run Tests
All changes saved
Any problems with the editor? Switch to basic editor
Give Feedback
Alka Rani
27°C Rain sho...
4:53 PM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Task 2
Java 8
by 2 to obtain 1;
• V = 1, which is odd: subtract 1 to obtain 0.
3. Given S = "1111010101111", the function should return 22.
4. Given string S consisting of "1" repeated 400,000 times, the function should return 799,999.
Write an efficient algorithm for the following assumptions:
• string S consists only of the characters '0' and/or '1';
• N, which is the length of string S, is an integer within the range [1..1,000,000];
• the binary representation is big-endian, i.e. the first character of string S corresponds to the most significant bit;
• the binary representation may contain leading zeros.
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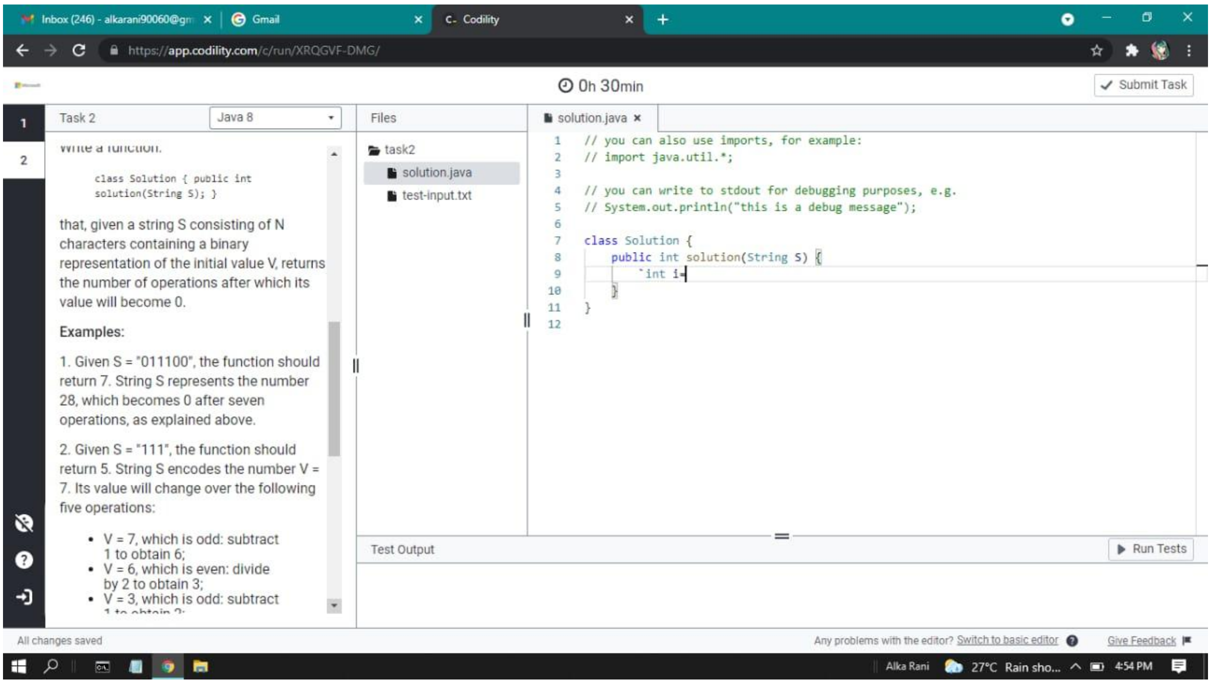
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Submit Task
Task 2
Java 8
Files
solution.java
Write a function:
class Solution { public int solution(String S); }
that, given a string S consisting of N characters containing a binary representation of the initial value V, returns the number of operations after which its value will become 0.
Examples:
1. Given S = "011100", the function should return 7. String S represents the number 28, which becomes 0 after seven operations, as explained above.
2. Given S = "111", the function should return 5. String S encodes the number V = 7. Its value will change over the following five operations:
V = 7, which is odd: subtract 1 to obtain 6;
V = 6, which is even: divide by 2 to obtain 3;
V = 3, which is odd: subtract
task2
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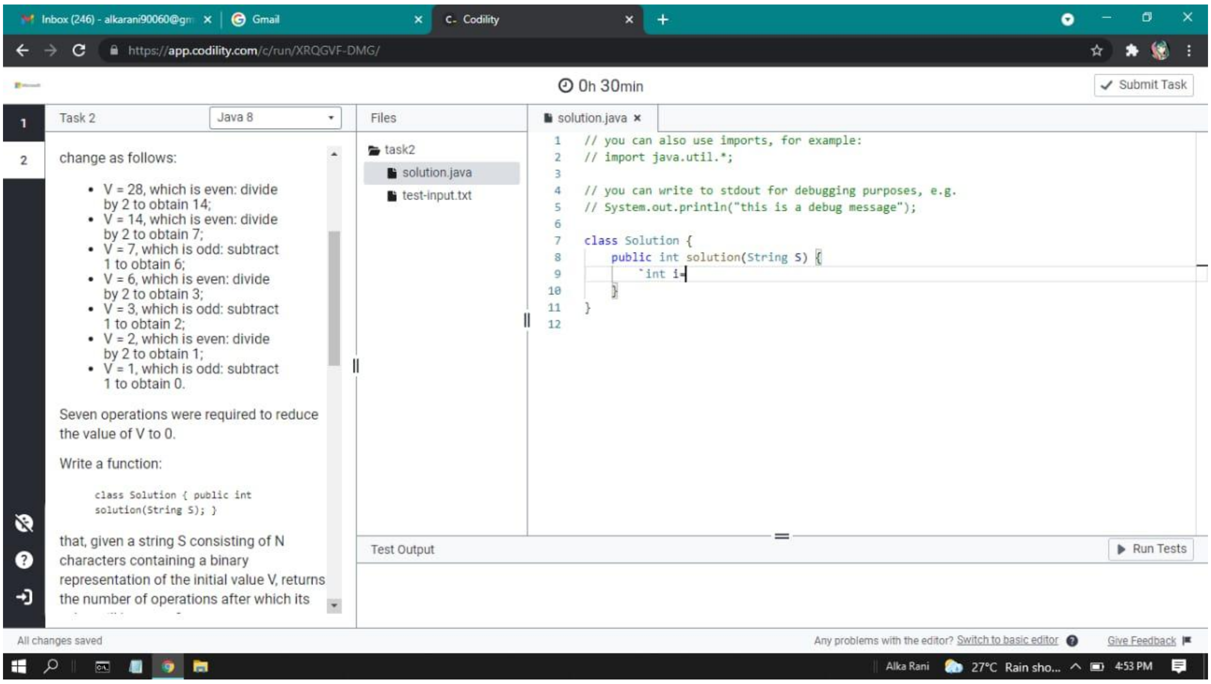
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0h 30min
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Task 2
Java 8
Files
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change as follows:
V = 28, which is even: divide by 2 to obtain 14;
V = 14, which is even: divide by 2 to obtain 7;
V = 7, which is odd: subtract 1 to obtain 6;
V = 6, which is even: divide by 2 to obtain 3;
V = 3, which is odd: subtract 1 to obtain 2;
V = 2, which is even: divide by 2 to obtain 1;
V = 1, which is odd: subtract 1 to obtain 0.
Seven operations were required to reduce the value of V to 0.
Write a function:
class Solution { public int solution(String S); }
that, given a string S consisting of N characters containing a binary representation of the initial value V, returns the number of operations after which its
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Solution:

Microsoft NIT Bhopal

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C++

Task 1

2

There is a road consisting of N segments, numbered from 0 to N-1, represented by a string S. Segment S[K] of the road may contain a pothole, denoted by a single uppercase "X" character, or may be a good segment without any potholes, denoted by a single dot, ".".

For example, string ".X..X" means that there are two potholes in total in the road: one is located in segment S[1] and one in segment S[4]. All other segments are good.

The road fixing machine can patch over three consecutive segments at once with asphalt and repair all the potholes located within each of these segments. Good or already repaired segments remain good after patching them.

Your task is to compute the minimum number of patches required to repair all the potholes in the road.

Write a function:

```
int solution(string &S);
```

that, given a string S of length N, returns the minimum number of patches required to repair all the potholes.

Examples:

1. Given S = ".X..X", your function should return 2. The road fixing machine could patch, for example, segments 0-2 and 2-4.

2. Given S = "X.XXXXX.X.", your function should return 3. The road fixing machine could patch, for example, segments 0-2, 3-5 and 6-8.

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https://app.codility.com/c/run/Z6P7VV-WDR/

Task 1 | C++

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3. Given S = "XX.XXX..", your function should return 2. The road fixing machine could patch, for example, segments 0-2 and 3-5.

4. Given S = "XXXX", your function should return 2. The road fixing machine could patch, for example, segments 0-2 and 1-3.

Write an efficient algorithm for the following assumptions:

- N is an integer within the range [3..100,000];
- string S consists only of the characters "." and/or "X".

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Task 1

C++

There is a road consisting of N segments, numbered from 0 to N-1, represented by a string S. Segment S[K] of the road may contain a pothole, denoted by a single uppercase "X" character, or may be a good segment without any potholes, denoted by a single dot, ".".

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Task 1

C++

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2. Given S = "X.XXXXX.X.", your function should return 3. The road fixing machine could patch, for example, segments 0-2, 3-5 and 6-8.

3. Given S = "XX.XXX..", your function should return 2. The road fixing machine could patch, for example, segments 0-2 and 3-5.

4. Given S = "XXXX", your function should return 2. The road fixing machine could patch, for example, segments 0-2 and 1-3.

Write an efficient algorithm for the following assumptions:

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1

2

Task 2

C++

that, given an array A consisting of N integers, returns the size of the largest possible subset of A such that AND of all its elements is greater than 0.

Examples:

1. Given A = [13, 7, 2, 8, 3] your function should return 3.

We can pick subset 13, 7 and 3. AND of these elements is positive and it is the largest possible subset of numbers that fulfills the criteria.

1101 (13) AND

0111 (7) AND

0011 (3) =

______

0001 (1)

2. Given A = [1, 2, 4, 8] your function should return 1. The AND of each pair from the array is equal to 0.

3. Given A = [16, 16] your function should return 2. The AND of both numbers is 16.

Write an efficient algorithm for the following assumptions:

- N is an integer within the range [1..100,000];
- each element of array A is an integer within the range [1..1,000,000,000].

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Task 2

C++

AND is a standard bitwise operation. For example, given K = 12 (binary representation 01100) and L = 21 (binary representation 10101) we obtain:

01100 AND

10101 =

______

00100

The AND operation can be extended to N integers, for example:

01100 AND

10101 AND

00100 =

______

00100

Because AND of 01100 (first argument) and 10101 (second argument) is 00100, and AND of this number with 00100 (third argument) is also 00100.

Write a function:

```
int solution(vector<int> &A);
```

that, given an array A consisting of N integers, returns the size of the largest possible subset of A such that AND of all its elements is greater than 0.

Examples: